What factors affect IMDB ratings above 7?

Part 1

Dataset

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and what the variables mean

Part 2

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before modelling

Part 3

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and the main results

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Dataset

- 2,847 films in the original dataset
- 7 main variables were used in this study
- IMDB rating was changed into a binary outcome: rating > 7
- Only film length has missing values (131 cases)

Table 1: Summary statistics for numerical variables

Variable	Min	1st Qu	Median	Mean	3rd Qu	Max	Missing
film_id	19.0	14159.5	29447.0	29180.86	43982.0	58748.0	0
year	1898.0	1957.0	1982.0	1975.75	1997.0	2005.0	0
length	1.0	73.0	90.0	82.22	101.0	480.0	131
budget	2.5	9.9	11.9	11.85	13.7	22.3	0
votes	5.0	11.0	29.0	657.03	114.0	149494.0	0
rating	0.8	3.7	4.6	5.34	7.7	9.2	0

Number of films by genre

genre	n
Action	872
Drama	797
Comedy	684
Animation	178
Documentary	154
Short	131
Romance	31

Action, Drama, and Comedy are the largest genre groups in the dataset, while Romance has the fewest films. This shows that the dataset is dominated by a few major genres, and some smaller genres may be less stable in later analysis.

Exploratory Analysis

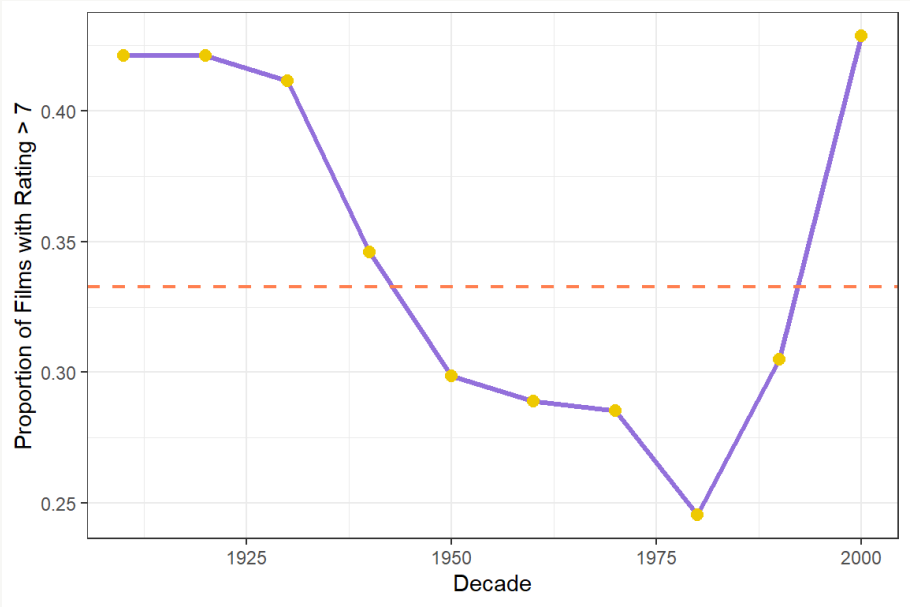


figure 1 the relationship between release year and the proportion of successful films

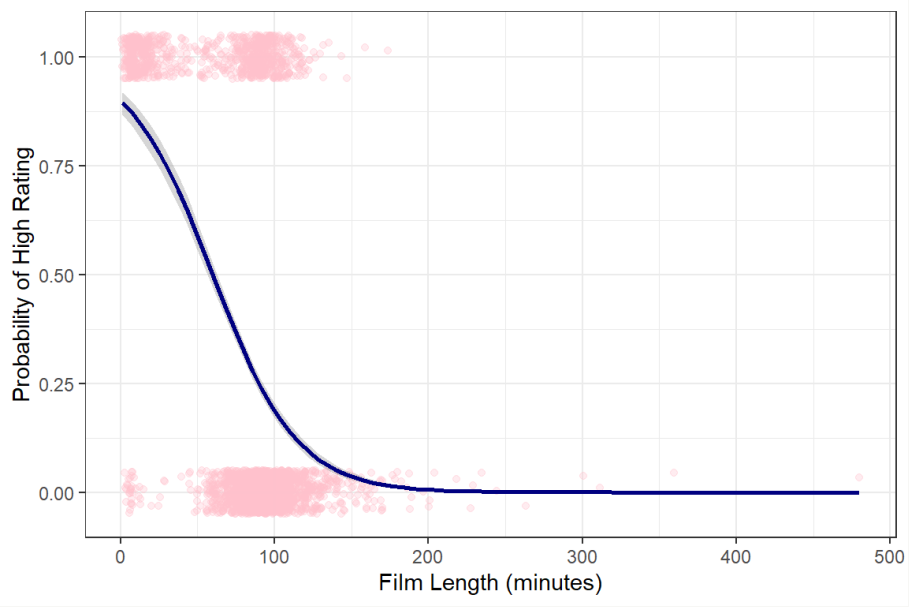


figure 2 the relationship between flim length and the proportion of successful films

The red line shows the average success rate. Success rate was above average before 1930 and around 2000, but below average from 1950 to 1990. One possible reason for the decline is that audience expectations increased with social development.

There is a clear downward trend. This means that shorter films are more likely to succeed than longer films. One possible reason is that shorter films are easier for audiences to understand and enjoy.

Exploratory Analysis

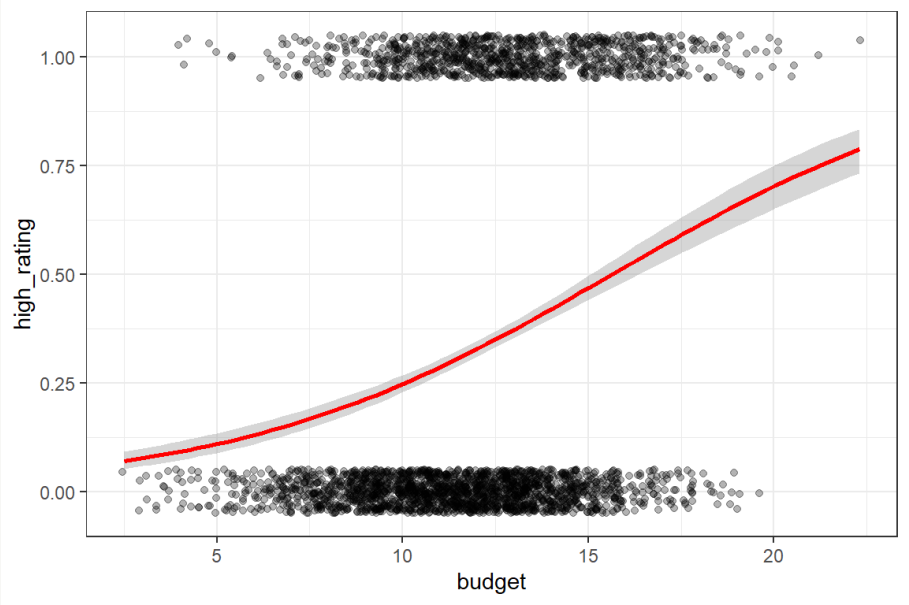


figure 3 the relationship between budget and the proportion of successful films

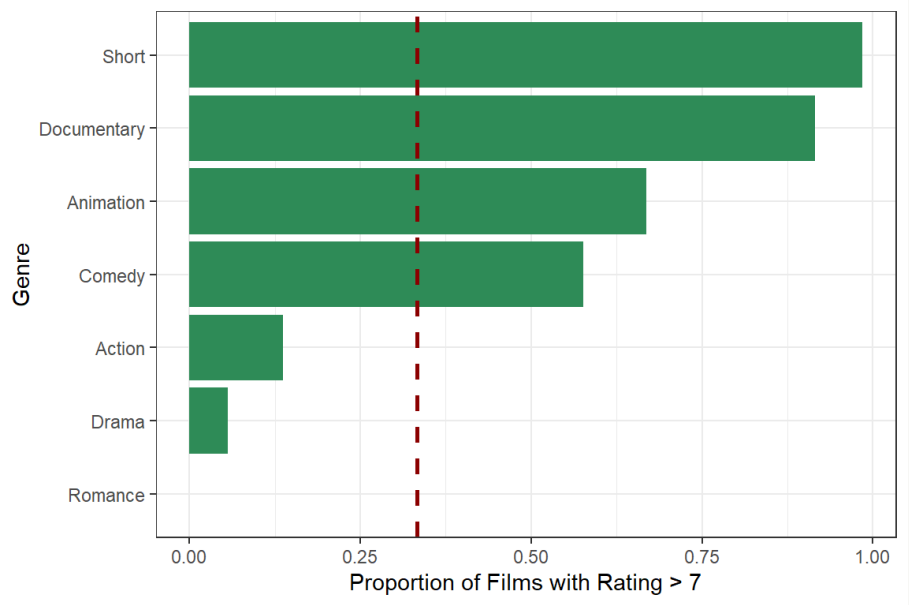


figure 4 the relationship between genre and the proportion of successful films

A clear upward trend can be seen in the figure 3. This means that as budget increases, the proportion of successful films also increases. In other words, films with higher budgets are more likely to be successful. One possible reason is that higher-budget films may have better production quality, stronger promotion, and wider audience attention.

The figure suggests that genre may affect film success. One possible reason is that audiences have different interests for different types of films, so some genres may be more likely to receive higher ratings than others.

Model Fitting

Full model

high_rating ~ year + length + budget + votes + genre

Variable	p value	Decision
year	0.001	keep
length	<0.001	keep
budget	<0.001	keep
votes	0.612	remove

Reduced model

high_rating ~ year + length + budget + genre

Main modelling result

- Votes was not statistically significant.
- Year, length, budget, and several genre effects stayed significant.
- The model identifies key drivers influencing a high IMDB rating.

Final model : high_rating ~ year + length + budget + genre

The reduced model achieved a **lower AIC** with **minimal changes** on **deviance**, providing a better balance between **model fit** and **simplicity** while retaining full explanatory power.

Table 8: Model fit statistics for the full and reduced models

Model	AIC	Residual_Deviance	DF
Full model	1478.365	1456.365	2705
Reduced model	1476.593	1456.593	2706

Note: 'high_rating' is binary; a Logit link function is used to model the probability of success.

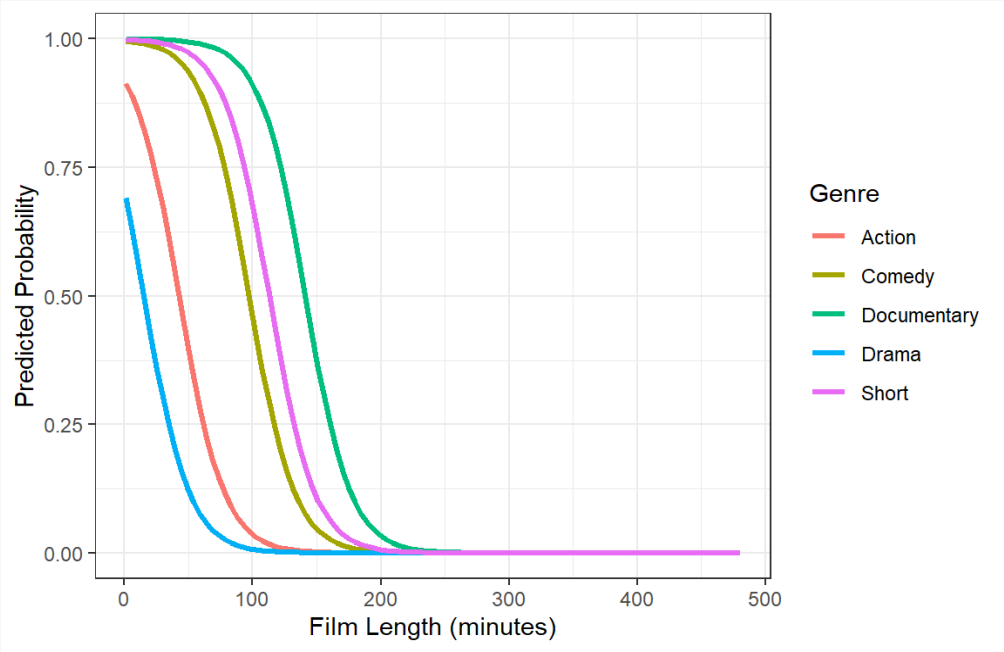
Final Model: Main Findings

Easy reading of the final model

- As year increases, the odds go up slightly (OR 1.010).
- As film length increases, the odds go down (OR 0.945).
- Higher budget raises the odds (OR 1.665).
- Documentary, Short, and Comedy have much higher odds than Action.
- Drama has lower odds than Action.

Table 9: Odds ratios for the reduced logistic regression model

Predictor	OR	SE	z	p	CI low	CI high
(Intercept)	0.000	5.767	-4.138	0.000	0.000	0.000
year	1.010	0.003	3.483	0.000	1.005	1.016
length	0.945	0.004	-16.077	0.000	0.938	0.951
budget	1.665	0.030	16.933	0.000	1.572	1.769
genreAnimation	0.924	0.320	-0.246	0.806	0.491	1.725
genreComedy	22.439	0.179	17.362	0.000	15.905	32.119
genreDocumentary	271.756	0.442	12.673	0.000	119.805	683.371
genreDrama	0.211	0.239	-6.509	0.000	0.130	0.333
genreRomance	0.000	391.829	-0.037	0.970	0.000	207.390
genreShort	54.903	0.797	5.028	0.000	14.173	370.971



What we learn

- As film length increases, the predicted probability of a high rating becomes lower.
- Documentary films keep the highest predicted probability across most film lengths.
- Drama films have the lowest predicted probability.
- The figure shows the joint effect of film length and genre, while other variables are held constant.

Model Limitation

Limited variables	The dataset does not include factors such as director, cast, awards, or marketing.
Missing values	Missing film length values were removed, which reduced the sample size and may cause bias.
Small genre groups	Some genres have few films, so their coefficient estimates may be less stable.

Future work & Conclusion

Future work

1. Expand variables: Add factors like director, cast, and awards to the model.
2. Handle missing data: Use imputation methods instead of removing incomplete records.
3. Explore interactions: Investigate interaction effects between variables like budget and release periods.

Key message

Longer films were less likely to receive high ratings, while newer films, higher-budget films, and some genres showed higher probabilities of receiving ratings above 7.

Data Source: A comprehensive subset of the IMDB database

Thank You